Fiscal Year:	FY 2015	**Task Last Updated:**	FY 04/16/2015
PI Name:	Nickerson, Cheryl A Ph.D.		
Project Title:	RNA Deep Sequencing and Metabolomic Profiling of Microgravity-Induced Regulation of the Host-Pathogen Interaction: An Integrated Systems Approach		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Immunology		
Space Biology Special Category:	(1) Cell Culture (2) Translational (Countermeasure) Potential		
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PI Organization Type:	UNIVERSITY	**Phone:**	480-727-7520
Organization Name:	Arizona State University		
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PI Web Page:	https://		
City:	Tempe	**State:**	AZ
Zip Code:	85287-5401	**Congressional District:**	9
Comments:	NOTE PI moved from Tulane University to Arizona State University in 2006.		
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/01/2013	**End Date:**	09/30/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	1	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	ISS		
Flight Assignment:	NOTE: Extended to 9/30/2017 per NSSC information (Ed., 6/29/17) NOTE: Extended to 6/30/2017 per F. Hernandez/ARC (Ed., 4/14/16)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Aronow, Bruce (Cincinnati Children's Hospital) Khalpey, Zain (University of Arizona) Ott, Mark (NASA Johnson Space Center) Sams, Clarence (NASA Johnson Space Center) Barrila, Jennifer (CoPI/ Arizona State University)		
Grant/Contract No.:	NNX13AM01G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	A major challenge in mitigating infectious disease risks during spaceflight is to understand, model, and integrate the combined action of cellular, molecular, and biochemical networks in the host that potentiate transition to disease in response to infection by pathogens cultured in microgravity. This is a critical issue to address, as spaceflight negatively impacts crew immune function and alters microbial virulence, gene expression, and antimicrobial resistance, strongly suggesting an increased risk for in-flight infectious disease. Herein, we propose to apply our 3-D intestinal co-culture models as human surrogates to characterize their transcriptomic (via RNA-Seq), metabolomic (via mass spectrometry), and inflammatory responses to infection with spaceflight analogue and control-cultured Salmonella typhimurium. To facilitate a practical integration of these systems, the combined analysis proposed in this study will focus on the metabolomic subset associated with host oxidative stress, redox, and inflammatory responses. These factors are of key importance in the infection process and may contribute to reported abnormalities and dysfunction in the crews' immune system during flight. We will use systems modeling approaches based on both quantitative gene transcript and biochemical metabolite correlation and all of their known/inferable relationships between genes, proteins, pathways, and biochemistry that will allow us to form new bridges and a general framework for making these linkages between gene expression (RNA-Seq) and biochemical (metabolomic) data. This approach will allow for new insights into relationships between redox functions, oxidative stress, inflammation, and infection that either approach alone would not be able to achieve. By exploring interconnections between these systems over different kinetic timepoints before and after infection, this systematic approach will provide an unparalleled level of sensitivity and resolution of the dynamics of the host response to a microgravity-analogue cultured pathogen, which may lead to identification of novel infection mechanisms and strategies for prevention and control.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research will enrich life on Earth through the use of space technology and the application of biomedical knowledge. Specifically, this study will utilize the microgravity spaceflight platform to 1) to broaden our knowledge of the host-pathogen interaction that leads to infectious disease, and 2) for the development of new therapeutic strategies to combat infectious disease for the general public.
Task Progress:	In vitro infection assays have been performed in biological triplicate using standard gentamicin protection protocols. The results revealed unique and statistically significant differences in the colonization profiles over a series of kinetic timepoints of the 3-D co-culture model infected with Salmonella grown under spaceflight analog conditions relative to the control. 3-D cells were monitored for viability under microenvironmental parameters relevant to those encountered by Salmonella in the parental tissue in vivo. In addition, all final samples have been collected for RNA-seq, cytokine, and metabolomic analyses.
Bibliography Type:	Description: (Last Updated: 07/02/2025)
Articles in Peer-reviewed Journals	Kennedy AR, Crucian B, Huff JL, Klein SL, Morens D, Murasko D, Nickerson CA, Sonnenfeld G. "Effects of sex and gender on adaptation to space: Immune system." J Womens Health (Larchmt). 2014 Nov;23(11):956-8. http://dx.doi.org/10.1089/jwh.2014.4913 ; PubMed PMID: 25401940; PubMed Central PMCID: PMC4236064 , Nov-2014
Articles in Peer-reviewed Journals	Crabbé A, Liu Y, Sarker SF, Bonenfant NR, Barrila J, Borg ZD, Lee JJ, Weiss DJ, Nickerson CA. "Recellularization of decellularized lung scaffolds is enhanced by dynamic suspension culture." PLoS One. 2015 May 11;10(5):e0126846. eCollection 2015. http://dx.doi.org/10.1371/journal.pone.0126846 ; PubMed PMID: 25962111; PubMed Central PMCID: PMC4427280 , May-2015